Age-for-Time Anchoring with Shift-Aligned Residual Learning for Global Land Ecosystem Forecasting under Future Climate Scenarios

Hansika N M, Jayasree B, Janani S P
Department of Computer Science and Engineering
SSN College of Engineering, Chennai, India

***Abstract*—Learned emulators of process-based ecosystem models are usually trained to roll an ecosystem state forward year by year and are judged on locations and climates that resemble the training data. The IEEE Big Data Cup 2026 CarbonGlobe challenge scores something different: the year-40 height and aboveground biomass (AGB) of 6,171 sites under two future climate scenarios (SSP126 and SSP585), where the test climate and the test initial states differ markedly from the training data. We show that in this setting a non-autoregressive formulation outperforms the sequence emulators we built first. Our method (i) aligns the test initial states to the global training distribution by per-age quantile mapping; (ii) forms an *age-for-time anchor*, the site's own initial-state curve read 40 years further along the seed-age axis and re-calibrated per age; (iii) learns only the anchor residual with gradient-boosted trees from eight summary statistics of each of 136 climate drivers, aligned to the training distribution by a pooled mean–scale transform that preserves the contrast between scenarios; and (iv) shrinks the residual with weights fitted on directional-extrapolation folds. On the public leaderboard (mean squared relative error, lower is better) the anchor alone scores 0.093 and the final model 0.066, against 0.153 for a CNN–BiGRU–GRU emulator with an age-baseline residual. Leaderboard probes built by swapping components between submissions show that the residual's entire gain is in AGB and that 89% of it comes from ages of 50 years and above, and they reject a uniform AGB-uplift hypothesis. We also report that random and cluster validation ranked our candidate models in the opposite order from the leaderboard, and we document the failed alternatives.**

***Index Terms*—CarbonGlobe, ecosystem emulation, distribution shift, quantile mapping, covariate alignment, residual learning, gradient boosting, space-for-time substitution, SSP scenarios**

I. Introduction

Process-based ecosystem models such as the Ecosystem Demography (ED) model simulate forest growth, mortality and carbon exchange from climate and soil forcing [4], but running them globally for many scenarios is expensive. Machine-learning emulators promise a cheap substitute. CarbonGlobe provides an ML-ready benchmark for this purpose, with 40 years of monthly environmental forcing, 136 input variables, 15 forest-age conditions and 54,152 land sites at 0.5° resolution, together with ED outputs for seven ecosystem variables [1]. The IEEE Big Data Cup 2026 challenge [2] asks for forecasts at 6,171 held-out sites under the low-emission SSP126 and high-emission SSP585 scenarios [5], but it scores only the year-40 height and AGB.

Two properties of this task are unusual. First, only the end state after 40 years is scored, so a model need not reproduce the intermediate trajectory. Second, the test inputs lie outside the training distribution: future climate is shifted relative to the present-day training climate, and the provided initial states of the test sites do not follow the same marginal distributions as the global training data (Section III). Yet the common emulator recipe is a year-by-year autoregressive rollout trained on random sites [1], [3]. In our own experiments such emulators reached validation errors near 0.02 on random held-out sites but 0.153 on the leaderboard.

The data also contain an axis that rollout models use only as an input: each site is simulated from 15 seed ages (1 to 500 years). For old stands the state changes slowly (mean height at age 50 rises by about 8% over the whole 40 years in a 2,000-site sample), so the site's own initial-state curve across ages already says what a stand of age $a+40$ looks like at that site. This is a chronosequence, or space-for-time, argument [6] applied along the age axis of a single site.

A. Novelty and Contributions

To the best of our knowledge, the combination below has not been reported for this benchmark. The contributions, in order of importance, are:

- **Age-for-time anchor.** A non-autoregressive predictor of the year-40 state from the site's own initial-state curve read at age $a + 40$ and re-calibrated per age, with no climate input. Alone it scores 0.093 on the public leaderboard, better than our neural emulators (0.153 to 0.156) and than the best autoregressive result we are aware of for this challenge (0.124) [3].
- **Two-stage shift alignment.** (i) Per-age, per-variable quantile alignment of the test initial states to the global distribution with random tie-breaking; (ii) a pooled mean–scale alignment of climate summaries, computed over both scenarios together so that the contrast between SSP126 and SSP585 is preserved. Keeping and aligning all 1,088 climate summaries scored 0.066, against 0.073 for discarding shifted drivers.
- **Shrunk residual learning on 54,152 sites.** Gradient-boosted trees learn only the anchor residual from 812,280 site–age sequences (16 times more sites than the regional split). Residual weights are fitted per target and age group

by non-negative least squares on directional-extrapolation folds and capped.

- **Leaderboard probe decomposition.** A protocol that attributes leaderboard error to targets and ages with two or three submissions made by swapping components between files, plus an analytic weight curve fitted from three scores. It shows that the residual helps AGB at ages ≥ 50 only, and it falsifies a plausible CO_2-style uplift hypothesis.
- **A documented validation failure.** Random, cluster and directional-extrapolation validation all ranked candidate models differently from the leaderboard. We report this with numbers (Section VI-D) because it determines how such challenges should be validated.

II. Related Work

A. Emulating process-based ecosystem models

CarbonGlobe's reference baselines include recurrent and attention-based forecasters (LSTM, LSTNet, DeepED, Transformer, Informer, DLinear, Crossformer and TimeXer) that predict multi-decadal trajectories from forcing and initial states [1]. Within this challenge, a peer study predicts annual state changes with a causal dilated temporal convolutional network (TCN) [12] and with a causal patch Transformer [13], reaching public scores of 0.142 and 0.124 [3]. These systems are autoregressive, are trained on the 3,373-site regional split, and treat seed age as a conditioning input. Our method differs in using the age axis as a predictor of the target itself and in training on the global dataset.

B. Space-for-time substitution

Ecologists infer temporal change from stands of different ages observed at one time [6]. We use the same idea inside one site: the simulated initial-state curve over 15 ages is a ready-made chronosequence, and reading it 40 years further along gives an estimate that needs no forcing at all.

C. Distribution shift and alignment

Predictive inference under covariate shift is classically handled by reweighting [8], [9]. Climate science routinely removes distributional bias with quantile mapping [7]. We apply quantile mapping to the initial states of the test sites and a mean–scale alignment to climate summaries, and we discuss why neither can be validated with labels (Section VIII).

D. Gradient boosting and shrinkage

Boosted trees [10] are strong on tabular data, saturate rather than extrapolate outside the training range, and train in minutes. Non-negative least squares [11] gives stacking weights that cannot flip sign; we additionally cap their sum.

III. Problem and Data

A. Task and metric

For site s and seed age a_j ($j = 1, \ldots, 15$; ages $1, 10, 20, 30, 50, 70, 100, 150, 200, 250, 300, 350, 400, 450, 500$), the input is an initial state $y^0_{s,j} \in \mathbb{R}^7$ (height, AGB, soil carbon, LAI, GPP, NPP, Rh) and a forcing tensor $X_s \in \mathbb{R}^{40\times12\times136}$. The output is $z_{s,j} = (h_{s,j}/\mu_h,\ b_{s,j}/\mu_b)$, the year-40 height and AGB divided by their global means $\mu_h = 10.90645$ and $\mu_b = 3.39567$. The score is

$$S = \tfrac{1}{2}\left(\mathrm{MSE}_h + \mathrm{MSE}_b\right), \tag{1}$$

the mean squared relative error; $\sqrt{S}$ is the relative RMSE. The test set has 6,171 sites, 15 ages and two scenarios (185,130 rows). Public and private leaderboards use a hidden random split of the sites.

TABLE I
Adversarial AUC (global vs. test, 15 ages per variable). 0.5 means indistinguishable.

Variable	Raw	After quantile alignment
Height	1.000	0.955
AGB	1.000	0.831
Soil carbon	1.000	0.873
LAI	0.991	0.854
GPP	0.996	0.797
NPP	0.996	0.861
Rh	0.996	0.681

B. Training data layout

The global training data hold, for each of 54,152 sites and each seed age, a $41 \times 12 \times 7$ array: years 0 to 40, with the 12 monthly slots repeated because targets are annual. We take y^0 from year 0 and the target from year 40, and use the climate tensor of the same site. A regional split of 3,373 training and 852 test sites with full yearly trajectories is also provided; our neural baselines use it.

C. What differs between training and test

Two shifts matter. *Climate:* 624 of the 1,088 climate summaries have a pooled test mean within 0.3 global standard deviations of the global mean; the others are shifted further. Along the global-to-test mean-shift direction the test is 9.06 units away, against a standard deviation of 3.08 for the global sites (Section IV-F).

Initial states: a classifier trained to separate global from test sites using the 15 age values of one variable reaches an AUC of 0.99 or higher for every variable on the raw values (Table I). The test values also have constant minima that the global values do not: negative for height, AGB and soil, positive for LAI, GPP, NPP and Rh. We could not identify the transformation that produced them. Quantile alignment (Section IV-B) reduces but does not remove the separability.

IV. Method

A. Overview

Fig. 1 and Algorithm 1 show the pipeline. The test initial states are aligned to the global distribution and give an anchor from the age axis. Gradient-boosted trees then predict the anchor's error from the aligned initial-state curve and aligned climate summaries. The final forecast is the anchor plus a capped weighted sum of residual predictions.

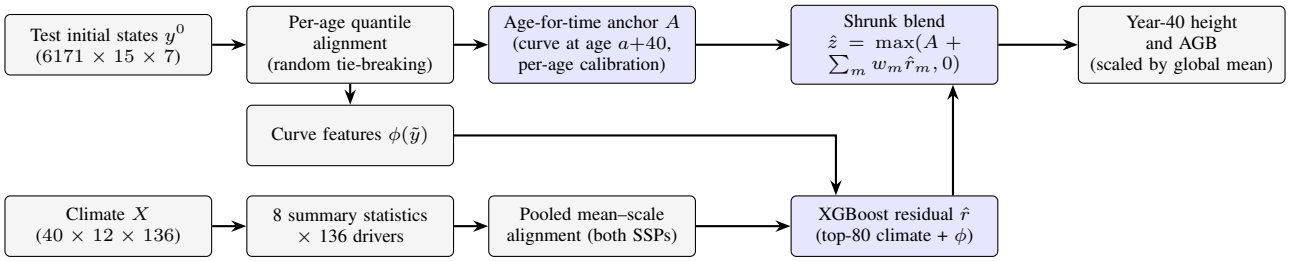


Fig. 1. Proposed pipeline. Shaded blue boxes are the components that carry the forecast: the anchor, the residual model and the shrunk blend.

Algorithm 1 Inference for one scenario pair

Require: test y^0 (shared by both SSPs), climate $X^{(126)}, X^{(585)}$; global statistics and trained models

$\tilde{y} \leftarrow$ QUANTILEALIGN(y^0) per age and variable
$A \leftarrow$ ANCHOR$(\tilde{y})$ (Eqs. 3–4)
$g^{(\cdot)} \leftarrow$ summary statistics of $X^{(\cdot)}$ (136 drivers × 8)
align $g^{(126)}, g^{(585)}$ jointly to the global mean and scale (Eq. 5)
for each scenario and each model $m \in \{\text{y0-only}, \text{climate}\}$ **do**
  $\hat{r}_m \leftarrow \text{XGBOOST}_m(\phi(\tilde{y}), g)$ averaged over seeds
end for
$\hat{z} \leftarrow \max\left(A + \sum_m w^{(m)}_{k,\gamma(j)} \hat{r}_m,\, 0\right)$
return $\hat{z}$

B. Quantile alignment of initial states

For each age j and variable k, the n test values are ranked with ties broken at random, converted to levels $q_i = (\text{rank}_i - \frac{1}{2})/n$, and replaced by the global quantile at that level:

$$\tilde{y}_{s,j,k} = Q^G_{j,k}(q_{s,j,k})\,, \tag{2}$$

where $Q^G_{j,k}$ is the empirical quantile function of the 54,152 global values. Random tie-breaking matters because many test sites share an identical constant value (the minimum); collapsing them onto one aligned value is both unlike the global data and a trivial cue for an adversary. The method assumes that, at each age, the test sites are a random sample of the global population, which we cannot verify.

C. Age-for-time anchor

Let $\tilde{y}_{s,\cdot,k}$ be the aligned curve of variable $k \in \{\text{height}, \text{AGB}\}$ over the 15 ages, scaled by μ_k. Its value at age $a_j + 40$ is obtained by linear interpolation in age, with the query clipped to $[1, 500]$:

$$\hat{c}_{s,j,k} = \text{interp}\big(a \mapsto \tilde{y}_{s,a,k}/\mu_k;\ \min(a_j + 40, 500)\big). \tag{3}$$

A per-age, per-target linear calibration is fitted on the global sites by least squares,

$$A_{s,j,k} = \alpha_{j,k}\, \hat{c}_{s,j,k} + \beta_{j,k}. \tag{4}$$

The anchor uses no climate information. Its role is to supply the part of the year-40 state that the initial-state curve already contains, so that the learned component only has to model a correction.

D. Climate summaries and pooled alignment

For each of the 136 drivers we compute eight site-level statistics from the 40×12 series: the mean, the standard deviation of annual means, the mean within-year standard deviation, the late-minus-early decade difference, the seasonal amplitude, the means of the warmest and of the coldest three months, and the last-decade mean, giving 1,088 features per site. For feature f let m^G_f, σ^G_f be the global mean and standard deviation, and m^P_f, σ^P_f the same quantities over the *pooled* test sites of both scenarios. Test features are aligned by

$$x'_f = (x_f - m^P_f)\,\text{clip}\big(\sigma^G_f/\sigma^P_f,\, 0.5,\, 2\big) + m^G_f. \tag{5}$$

Pooling over both scenarios removes the shared shift while the difference between the SSP126 and SSP585 versions of a site survives. Features with no variation across global sites are dropped. The alternative we compare against keeps only features whose pooled test mean lies within 0.3 global standard deviations of the global mean.

E. Residual learning

The target of the learned model is $R_{s,j,k} = z_{s,j,k} - A_{s,j,k}$, trained on all 812,280 global site–age rows. The row features ϕ are the aligned values of the retained initial-state variables (height, AGB, GPP and Rh; the others had aligned AUC above 0.85 and were excluded as model inputs), their aligned curve values at ages a_j+20, a_j+40, a_j+60 and a_j+100, differences between these, the age and its logarithm. Two models are trained per target: m=1 uses ϕ only and m=2 adds the 80 most important aligned climate features, selected by gradient-boosting gain on site-level residual summaries after removing features correlated above 0.97 with a selected one. Gaussian noise of 0.08 standard deviations is added to the initial states of half of the training sites so that the models tolerate imperfect alignment. Each model is an XGBoost regressor [10] (learning rate 0.06, depth 6, minimum child weight 100, subsample 0.7, column sampling 0.5, ℓ_2 penalty 30, about 2,750 rounds, two seeds), trained on one NVIDIA T4 GPU; the full pipeline runs in under 40 minutes.

F. Shrunk blend and directional extrapolation folds

The final forecast is

$$\hat{z}_{s,j,k} = \max\Big(A_{s,j,k} + \sum_m w^{(m)}_{k,\gamma(j)}\, \hat{r}^{(m)}_{s,j,k},\, 0\Big), \tag{6}$$

where $\gamma(j)$ maps the age to one of three groups (1–30, 50–250, 300–500). The weights solve a non-negative least-squares problem on out-of-fold residuals [11] and are rescaled so that $\sum_m w^{(m)} \leq 0.7$.

Because a random split does not represent the climate shift, the folds are directional. Let d be the vector of standardised differences between the pooled test mean and the global mean of the 136 annual-mean drivers, $\hat{d} = d/\|d\|$, and u_s the projection of site s's standardised means on $\hat{d}$. We hold out the sites with the largest u_s ("up"), the smallest ("down"), and a random reference fold, choosing the held-out fraction so that the gap in mean u approaches $\|d\|$. Here $\|d\| = 9.06$ against a global standard deviation of u of 3.08, so the smallest allowed fraction of 5% (2,708 sites per directional fold) is used and the real shift is still larger than the folds simulate. Held-out initial states are evaluated both clean and with added noise of 0.10 standard deviations.

V. Experimental Protocol

A. Leaderboard use

Final comparisons use the public leaderboard (Eq. 1). Every number in Section VI is a public score unless it is labelled as validation. We disclose which choices were informed by it: the cap of 0.7 on the residual weight was set after seeing the 0.5 and 1.0 results of an earlier pipeline, and the final candidate was chosen among four submissions.

B. Probe decomposition

Eq. 1 is a sum over targets and rows. Let A be an anchor-only file and C the full model. A file that takes the residual for height and the anchor for AGB scores $e_h(C) + e_b(A)$; because $e_h(A) + e_b(A)$ and $e_h(C) + e_b(C)$ are known, one such submission fixes all four terms. Replacing the target split by an age split does the same for ages. A file $A + \lambda(C - A)$ gives a point on the weight curve; three points determine a parabola, so we use the curve only to locate its minimum, not to estimate uncertainty. The decomposition assumes that the leaderboard weights all rows equally, which holds for an unweighted MSE.

C. Neural baselines

For comparison we trained a sequence emulator on the 3,373-site regional split. It compresses the 12 monthly drivers of each year with a 1-D CNN, reads the 40 years with a bidirectional GRU [14], and predicts the year-40 state both directly and by a GRU-cell decoder that rolls forward yearly. Both heads predict a correction to a per-age linear baseline (the age mean plus an age slope times the year). Drivers are reduced to the 8 most shifted and the 15 most predictive, with the remaining ones compressed to 20 principal components; height and AGB receive three times the loss weight, and three seeds are averaged. An earlier version without the age baseline is denoted v8.

TABLE II
Public leaderboard scores of the main submissions (lower is better).

Submission	Score
v8: CNN–BiGRU–GRU emulator	0.156
v9: + per-age baseline residual	0.153
Anchor + 0.5 × boosted residual (first global pipeline)	0.074
Filtered climate, residual weight 1.0	0.089
Filtered climate, residual weight 0.5	0.068
Anchor only (final pipeline)	0.093
Filtered climate, capped weights	0.073
Average of filtered and aligned residuals	0.068
Aligned climate, capped weights (final)	**0.066**

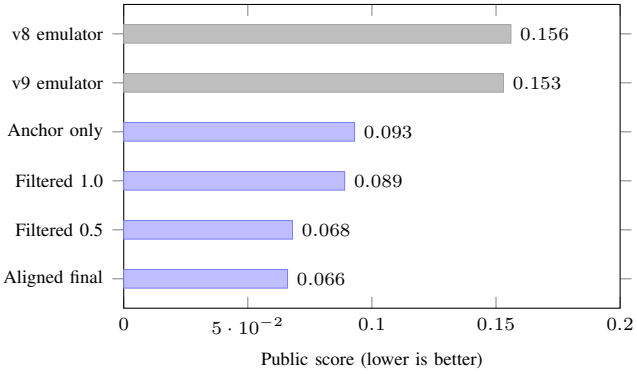


Fig. 2. Public scores of the neural emulators (grey) and of the anchor-based pipeline (blue). "Filtered" and "Aligned" name the climate treatment; the number is the residual weight for the filtered runs.

VI. Results

A. Progression of public scores

Table II and Fig. 2 list the main submissions. The neural emulators scored 0.156 and 0.153. Switching to the global data with an anchor and a boosted residual (full residual weight of 0.5 on an earlier anchor) gave 0.074. Residual models on shift-filtered climate scored 0.068 at half weight and 0.089 at full weight. In the final pipeline the anchor alone scored 0.093; capped residuals on filtered climate scored 0.073; aligned climate scored 0.066; and averaging the filtered and aligned residuals scored 0.068.

B. Where the residual helps

Table III and Fig. 3 show the probe submissions built from the anchor file A (0.093) and the final file C (0.066). Applying the residual to height only scores 0.094, slightly worse than the anchor, so the full gain of 0.027 is a gain in AGB (0.028 on AGB, -0.001 on height under the additive decomposition). Applying it only at ages of 50 years and above scores 0.069, which recovers 0.024 of the 0.027 (89%). Scaling the residual of C by 0.75 leaves the score at 0.066, which places the minimum of the weight curve between the two weights and the attainable score at about 0.066 for this residual.

TABLE III
PROBE SUBMISSIONS (PUBLIC SCORE). ALL START FROM THE ANCHOR A OR THE FINAL FILE C.

Submission	Score
A: anchor only	0.093
C: anchor + aligned-climate residual	0.066
C residual scaled by 0.75	0.066
Residual for height only, AGB = anchor	0.094
Residual for ages ≥ 50 only	0.069
C with AGB $\times(1+0.04)$ / $\times(1+0.08)$ (SSP126 / SSP585)	0.075
C with AGB $\times(1+0.08)$ / $\times(1+0.16)$ (SSP126 / SSP585)	0.097

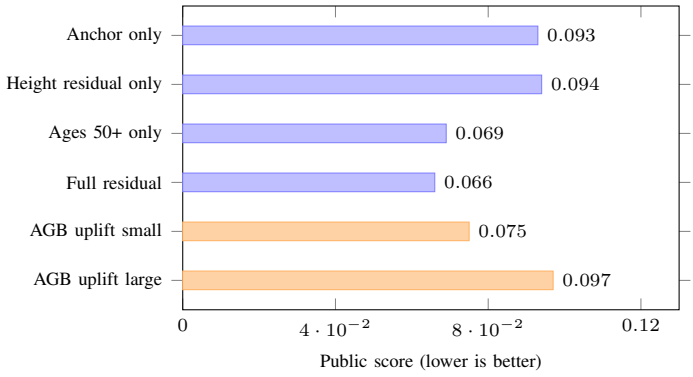


Fig. 3. Probe decomposition. Blue: where the residual is applied. Orange: uniform AGB uplift applied on top of the final file (the larger uplift is twice the smaller).

The uplift probes test a specific hypothesis: that AGB is systematically too low under the future climates, for instance because CO_2, a forcing variable, lies far outside the training range, so the error would shrink if AGB were scaled up more for SSP585 than for SSP126. Both uplifts raise the error. A parabola through the three points places the best uplift at about -1.5%, that is, at none. The remaining error is therefore not a global bias in AGB but a site-to-site error.

C. Weight curve

For the earlier filtered-climate residual the three scores (anchor 0.093, weight 0.5 giving 0.068, weight 1.0 giving 0.089) fit $S(\lambda) = 0.093 - 0.096\lambda + 0.092\lambda^2$, with a minimum of 0.068 at $\lambda \approx 0.52$. The full-weight submission was thus past the optimum by about a factor of two, and no weight could have taken that residual below 0.068. The aligned-climate residual is better: its attainable score is about 0.066.

D. Validation versus the leaderboard

Table IV lists validation errors beside public scores. For the first pipeline the mapped-input validation ranked the full-weight blend first (0.0595) and the half-weight blend third (0.0886); the leaderboard reversed this order (0.089 against 0.068). Directional-extrapolation folds for the final pipeline gave 0.034 to 0.041 for the capped blend (clean inputs), against a public score of 0.073 for the same model. On clean

TABLE IV
VALIDATION ERROR (MEAN SQUARED RELATIVE ERROR ON HELD-OUT SITES) VERSUS PUBLIC SCORE.

Model	Validation	Public
First pipeline, weight 1.0	0.060 (mapped)	0.089
First pipeline, weight 0.5	0.089 (mapped)	0.068
Final pipeline, filtered, capped	0.034 (up fold, clean)	0.073
Anchor only (up / down fold)	0.100 / 0.114	0.093

TABLE V
POSITIONING AGAINST AUTOREGRESSIVE EMULATORS [1], [3].

Aspect	Autoregressive emulators	This work
Formulation	Year-by-year rollout of 7 variables	One-step anchor plus residual for year 40
Age axis	Conditioning input	Chronosequence anchor
Training sites	3,373 regional	54,152 global
Shift handling	Input normalisation	Quantile + pooled alignment, shrinkage
Validation	Random sites, rollout	Directional folds and leaderboard probes
Output	Full trajectory	Year-40 height and AGB only

directional folds the anchor alone scored 0.100 and 0.114, and the best residual weight was still 1.0 on those folds, whereas the leaderboard preferred about 0.5 to 0.6. The anchor error on the test set (0.093) is lower than on any global fold (0.100 to 0.133), while the residual removes roughly two thirds or more of the anchor error on global folds but only about 29% on the test set (0.093 to 0.066).

E. Climate treatment

Replacing shift-filtered climate (the filter keeps 624 of 1,088 features at the 0.3 threshold) by all non-degenerate features aligned with Eq. 5 lowers the public score from 0.073 to 0.066. The two residual models disagree on the test set with a mean squared difference of 0.015, which is as large as their disagreement on the shifted validation folds. Averaging them gives 0.068, between the two.

VII. NOVELTY ANALYSIS

Table V contrasts our approach with the autoregressive emulators used for this benchmark. We separate what is new here from what is borrowed.

New. (i) Using the seed-age axis as a within-site chronosequence anchor for a 40-year forecast; (ii) a pooled alignment of climate summaries across scenarios that preserves scenario contrast; (iii) the swap-based probe decomposition, which turns a scalar leaderboard into target-wise and age-wise attributions; (iv) the quantitative record of the validation–leaderboard gap.

Borrowed. Quantile mapping [7], chronosequence reasoning [6], gradient boosting [10] and non-negative stacking [11]. Our claim is the combination and its measured effect, not the components.

Not claimed. The method does not produce the intermediate trajectory and is not an emulator of ED dynamics; it answers the scored question. It is not shown to generalise beyond this benchmark.

VIII. Threats to Validity

Leaderboard selection. The weight cap and the choice of the final submission used public scores, so 0.066 is mildly optimistic for the private split; we report all candidates (four final submissions) to make the selection visible. **No repeated seeds.** Each score is one run, so differences of 0.002 to 0.003 (for example 0.066 against 0.068) are within what seed variation could produce. **Alignment cannot be validated.** The alignment steps assume that test sites are a random sample of the global population at each age; no labelled data exist to check this for future climate, and the residual AUC after alignment (Table I) shows that the initial states still differ. **Additivity.** The probe decomposition assumes an unweighted mean over rows. **Hypotheses not tested.** We prepared offset-and-scale corrections of the initial states as an alternative to quantile mapping but have no score for them, so the choice of quantile mapping is not isolated. **Simulator target.** The labels are ED outputs; agreement with ED is not agreement with field data.

IX. Limitations and Future Work

The anchor presumes that the initial-state curve is informative, which holds for slowly changing old stands but less for young ones; the residual gain at ages below 50 is small (0.003 of 0.027). The height residual contributed nothing, so our height forecast is essentially the anchor. Future work should (i) resolve the mismatch between test and global initial states, for which the constant minima in the test values are a lead; (ii) use the full 41-year trajectories of all 54,152 sites to train a dynamics model on top of the anchor; (iii) construct a validation that tracks the leaderboard, for instance with shift magnitudes calibrated to the test set; (iv) combine a sequence emulator trained on the global data with the boosted residual; and (v) attribute the remaining error by age and target with further probes.

X. Conclusion

For the year-40 forecasting task of the CarbonGlobe challenge, predicting a small, shrunk correction to an age-for-time anchor outperformed the autoregressive and sequence emulators we built, reducing the public score from 0.153 to 0.066 (relative RMSE from about 39% to 26%). The decisive ingredients were the anchor, training on the full global dataset, aligning rather than discarding shifted climate features, and shrinking the residual. Leaderboard probes localised the gain to AGB at ages of 50 years and above and rejected a uniform uplift. Because validation did not predict the leaderboard, we regard validation under climate shift as the main open problem.

Acknowledgment

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